Syed Muhammad Osama

Electrical Engineer

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Work Permit Available

Skills

MATLAB / Simulink: ●●●●
LabVIEW / LabVIEW FPGA: ●●●○
Python: ●●●○
C / C++: ●●○○
PCB design (Altium Designer): ●●○○
Git: ●●○○
Microcontroller PIC / STM programming: ●●○○
Interpersonal Skills: ●●●●
Presentation: ●●●○
Agile Environment: ●●●○
Adaptability: ●●●●
Teamwork: ●●●●

Professional Experience

Working Student R&D Hardware ITE *(WSAudiology)* *07/2024 - Present*

- Implemented mannequin rotor control for Anechoic Chamber 2D/3D Polar measurements and EMI/EMC testing
- Performing software and hardware quality assurance tests for the hearing aids with UPL/UPV hardware
- Performing auxiliary data automation tasks for increased efficiency through XML scripting
- Performing necessary hearing aids data analysis through MATLAB

Avionics and Mission Systems Engineer *(Turkish Aerospace Industries)* *09/2022 - 10/2023*

- Implemented Fixed-Wing aircraft control using Pixhawk 6X Flight Controller with PX4 Autopilot
- Interfaced Air Data boom (RS485), Engine Control Unit (RS232), potentiometer actuator feedback (Analog through SPI), GPS, telemetry and First-Person View (FPV) system (UART)
- Tested aircraft control systems with Flight Simulator, using MATLAB/Simulink under different aircraft, aerodynamic data and environment parameters

Control Systems Engineer *(Public Office)* *08/2021 - 08/2022*

- Worked on flight data analysis of UAVs, for fault detection (actuator flutter or lagging response), implementing corrective strategies through controller gain tuning and optimization
- Tasked with actuator model estimation, control design and validation with embedded code of UAV control surfaces
- PIC Microcontroller C embedded programming for HILS testing to validate actuator control with real-time data logging

System Integration Engineer *(DAT FZCo. affiliated with Emerson - NI)* *09/2020 - 07/2021*

- Developed product-based solutions and integrated third-party test instruments with GUI using NI (now Emerson) instrumentation and software, primarily LabVIEW, LabVIEW FPGA as well as MATLAB
- Assembled remote test benches with industry-focused solutions through remote demos:
  1. Actuator and AC/DC motor control test bench
  2. Cable loom and wire harnessing test bench
  3. Temperature and Humidity chamber remote control
- Hands-on with Communication Protocol and interfaces (RS232, RS485, RS422, TTL. UART, CAN, SPI etc.)

Projects

- **Parametric Analyzer:** LabVIEW-based real-time monitoring and data logging. Post-flight analysis - using NI-CRIO-9046, integrating high quality sensors such as accelerometers, AHRS, pressure transducers, thermistors; calculating Euler angles, body rates, estimating Angle of Attack, and fault status updates
- **Shelter Monitoring System:** Truck shelter monitoring with custom RIO-based PCB design, interfacing different analog & serial communication digital sensors. Measurements included generator health, fuel level, wind speed and direction, cabin temperature & humidity, fuel tank and tires pressure

- **Power Amplifier Testing:** Power Amplifier Testing at different frequency bands using SDRs and PXI Signal Generator. Operated through programmable power supplies for signal interference system with complete wiring diagram using MS Visio and I^2C-based programming to control output based on power & health monitoring functionality
- **Center of Gravity Estimation:** Calculating trucks' CoG and horizontal alignment through axle loads via loadcells (pressure transducers) and hydraulic jacks using cDAQ & strain input module

Certifications

- **NI-CLAD**
  Certified LabVIEW Associate Developer - *Dec. 2020*
- **Workshop: Robust Control Design in MATLAB**
  Institute of Space Technology (IST) - *Mar. 2022*

Education

MSc Electromobility-ACES *FAU Erlangen-Nürnberg* *10/2023 - Present*
Relevant Courses: Artificial Intelligence, Electric Powertrain, Control Systems, Robotics, Programming & Mechatronics.
Project Thesis: Side-slip Estimation for Truck-Trailer Systems using Recurrent Neural Networks
Master Thesis: Comparison between Relevance Vector Regression and Gaussian Processes for System Identification

B.Sc. Electrical Engineering *SEECS, NUST H-12* *09/2016 - 07/2020*
Relevant Courses: Control Systems, Power Engineering, Electronics, Coding, RF systems
CGPA: 3.44/4.00
Thesis: Development of Automation of Canal Gate System
Actual gate motor design with analog control and implementation with centralized server-based On-Off control
Prototyping using LabVIEW, NI USB-6216 DAQ model

Languages

- **English** [Native]
- **German** [Basic] - A2